

Sanjay J K

Software Engineer — AI Systems Developer
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SUMMARY

Computer Science student at VIT (CGPA 9.34) building scalable backend systems and AI-powered applications, with experience in event-driven architectures, database-backed software, secure data pipelines, and Retrieval-Augmented Generation (RAG). Published two Indian patent applications in applied AI and co-authored an accepted international conference paper.

SKILLS

Languages: Python, TypeScript, JavaScript, C++, Java, C

Systems & Backend: Event-Driven Architecture, REST APIs, WebSockets, OAuth2, Concurrency, Docker, Redis, PostgreSQL, MongoDB, SQL, ChromaDB

AI / ML: LLM Pipelines, Retrieval-Augmented Generation (RAG), Agentic AI, Prompt Engineering, Model Evaluation, NLP

Frameworks & Web: React, Next.js, FastAPI, Node.js, Express.js, TailwindCSS

Developer Tools: Git, GitHub, LangChain, Langfuse, Ollama

EDUCATION

Vellore Institute of Technology (VIT) — B.Tech Computer Science and Engineering

2023 – 2027

CGPA: **9.34**

PROJECTS

Retrieval-Augmented Generation System

Python, LangChain, ChromaDB

- Engineered a hybrid RAG pipeline combining chunking, MiniLM embeddings, BM25, dense vector search via Reciprocal Rank Fusion, and cross-encoder re-ranking to improve retrieval precision by **31%** over a vector-only baseline.
- Built citation-confidence gating and an evaluation pipeline using Recall@K, MRR, NDCG, and Langfuse tracing to benchmark retrieval quality and keep answers grounded in source documents.

InterviewAI — Real-Time Voice AI Interviewer

FastAPI, React, WebSockets, Ollama

- Built a real-time AI interviewer using an event-driven full-duplex WebSocket pipeline integrating Deepgram ASR, local LLM inference, and ElevenLabs TTS.
- Developed a reproducible benchmark harness that reports speech-end-to-first-token, speech-end-to-first-audio, and completion timings, retains failures/timeouts, and surfaces event-loop scheduling delay through structured traces and dashboard telemetry.

NextUp — AI-Powered Placement Intelligence Platform

Next.js, FastAPI, PostgreSQL, Redis

- Built a placement intelligence platform that ingests CDC emails, extracts structured drive details, detects shortlists, and checks eligibility using LLM extraction backed by deterministic grounding gates.
- Designed a zero-knowledge security model using browser-side AES-256-GCM encryption and peppered HMAC blind indexes, and optimised a memory-constrained backend by eliminating an N+1 query bottleneck and adding circuit-breaker failover.

PATENTS & PUBLICATIONS

- Published Indian Patent Application (Indian Patent Office, 2026):** Personalized AI recommendation system for colour vision deficiency using perceptual gamut modelling.
- Published Indian Patent Application (Indian Patent Office, 2026):** Vision-language model for restoration and cross-era Tamil script decipherment.
- SET2026 Conference (Accepted, 2026):** Co-authored research on a physics-informed machine learning framework for thermosyphon heat-pipe design, contributing to data processing, predictive modelling, and validation of experimental systems.

EXPERIENCE

International Research Scholar — UKIERI-SPARC Program, Heriot-Watt University

Jun 2025

- Selected as **1 of 9 students nationally** for an international hydrogen-energy research exchange, conducting catalyst validation experiments and analysing results using cyclic voltammetry and linear sweep voltammetry.
- Co-presented a research-to-startup roadmap covering technical findings, budget, and scalability, translating experimental data into a structured business case.

Frontend Developer Intern — Valsco Technology (Remote)

Jun 2024 – Sep 2024

- Scaled **NoteSwap** to **200+ active student users** by developing a component-based React frontend with reusable UI modules and REST API integration, cutting page load time by **20%** through lazy loading and render optimization.
- Improved cross-device usability by designing a responsive component library adopted across the full desktop and mobile interface.

AWARDS & RECOGNITION

- GATE CSE 2026: AIR 1603 (Score: 670).**
- Judges' Choice Award – Sankalp Innovation Challenge 2026** (MNNIT Allahabad) for **JanVedha AI**, an explainable AI platform for civic governance and decision support.
- Finalist – Google Cloud Agentic AI Hackathon 2025 & India Innovates 2026; Top 3 – DeepShiva LLM Hackathon.**
- Board Member (Design Lead), AEE VIT; Senior Core Member, CodeChef VIT & TAM-VIT.**